

|                                           |                                                                                   |        |                                                                                   |        |                                                                                   |        |                                                                                     |        |                                                                                     |        |
|-------------------------------------------|-----------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------|--------|
| Single Band Routers                       |  |        |  |        |  |        |  |        |  |        |
|                                           | ASUS                                                                              |        | ASUS                                                                              |        | Digisol                                                                           |        | Digisol                                                                             |        | Digisol                                                                             |        |
| Brand                                     | ASUS                                                                              |        | ASUS                                                                              |        | Digisol                                                                           |        | Digisol                                                                             |        | Digisol                                                                             |        |
| Model no.                                 | RT-N12HP                                                                          |        | RT-N10E                                                                           |        | DG-BR4313NG                                                                       |        | DG-BR4000N/E                                                                        |        | DG-BG4100N                                                                          |        |
| Price                                     | 4,500                                                                             |        | 1,300                                                                             |        | 2,300                                                                             |        | 1,300                                                                               |        | 2,000                                                                               |        |
| Features                                  |                                                                                   |        |                                                                                   |        |                                                                                   |        |                                                                                     |        |                                                                                     |        |
| Total LAN ports/ Gigabit ports            | 5 / 0                                                                             |        | 5 / 0                                                                             |        | 5 / 0                                                                             |        | 5 / 0                                                                               |        | 4 / 0                                                                               |        |
| Antennae                                  | External                                                                          |        | External                                                                          |        | External                                                                          |        | External                                                                            |        | External                                                                            |        |
| No. of Antennae                           | 2                                                                                 |        | 1                                                                                 |        | 2                                                                                 |        | 1                                                                                   |        | 1                                                                                   |        |
| Antenna gain (dBi)                        | 9                                                                                 |        | 5                                                                                 |        | 3                                                                                 |        | 3                                                                                   |        | 2                                                                                   |        |
| Total USB/ USB 3.0                        | 0 / 0                                                                             |        | 0 / 0                                                                             |        | 1 / 0                                                                             |        | 0 / 0                                                                               |        | 0 / 0                                                                               |        |
| Data Rate (Mbps)                          | 300                                                                               |        | 150                                                                               |        | 300                                                                               |        | 150                                                                                 |        | 150                                                                                 |        |
| Wi-Fi Hardware Switch                     | N                                                                                 |        | N                                                                                 |        | Y                                                                                 |        | N                                                                                   |        | N                                                                                   |        |
| Inbuilt ADSL2+ modem                      | N                                                                                 |        | N                                                                                 |        | N                                                                                 |        | N                                                                                   |        | Y                                                                                   |        |
| Multiple SSIDs / Guest                    | N                                                                                 |        | N                                                                                 |        | N                                                                                 |        | N                                                                                   |        | Y                                                                                   |        |
| Parental / Access Controls                | Y                                                                                 |        | N                                                                                 |        | Y                                                                                 |        | Y                                                                                   |        | Y                                                                                   |        |
| UPnP                                      | Y                                                                                 |        | Y                                                                                 |        | Y                                                                                 |        | Y                                                                                   |        | Y                                                                                   |        |
| VPN Support                               | Y                                                                                 |        | N                                                                                 |        | N                                                                                 |        | Y                                                                                   |        | Y                                                                                   |        |
| PPPoE support                             | N                                                                                 |        | Y                                                                                 |        | Y                                                                                 |        | Y                                                                                   |        | Y                                                                                   |        |
| WPS Button                                | Y                                                                                 |        | Y                                                                                 |        | Y                                                                                 |        | Y                                                                                   |        | Y                                                                                   |        |
| Warranty (years)                          | 3                                                                                 |        | 3                                                                                 |        | 5                                                                                 |        | 5                                                                                   |        | 5                                                                                   |        |
| DD-WRT/Tomato support                     | N                                                                                 |        | N                                                                                 |        | N                                                                                 |        | N                                                                                   |        | N                                                                                   |        |
| Range Extender (WDS)                      | Y                                                                                 |        | Y                                                                                 |        | N                                                                                 |        | Y                                                                                   |        | N                                                                                   |        |
| Performance Tests                         | Zone 1                                                                            | Zone 2 | Zone 1                                                                            | Zone 2 | Zone 1                                                                            | Zone 2 | Zone 1                                                                              | Zone 2 | Zone 1                                                                              | Zone 2 |
| Throughput test (2.4 GHz) (mbps)          |                                                                                   |        |                                                                                   |        |                                                                                   |        |                                                                                     |        |                                                                                     |        |
| TCP Down                                  | 53.29                                                                             | 41.4   | 4.99                                                                              | 4.44   | 52.79                                                                             | 31.3   | 62.52                                                                               | 29.5   | 38.95                                                                               | 54.3   |
| TCP UP                                    | 88.94                                                                             | 64     | 7.79                                                                              | 34.4   | 69.76                                                                             | 39.6   | 72.59                                                                               | 17.3   | 50.34                                                                               | 48.7   |
| UDP Down                                  | 92.94                                                                             | 10     | 67.9                                                                              | 10     | 90.86                                                                             | 10     | 91.23                                                                               | 10     | 51.16                                                                               | 9.98   |
| UDP Up                                    | 95.08                                                                             | 10     | 3.61                                                                              | 10     | 66.11                                                                             | 9.21   | 67.84                                                                               | 10     | 46.39                                                                               | 10     |
| Steaming HD video (megabits/second)       |                                                                                   |        |                                                                                   |        |                                                                                   |        |                                                                                     |        |                                                                                     |        |
|                                           | 13.14                                                                             | 12.58  | 12.28                                                                             | 12.32  | 14.84                                                                             | 12.83  | 12.98                                                                               | 4.87   | 12.81                                                                               | 13.16  |
| Signal Strength (dBi)                     |                                                                                   |        |                                                                                   |        |                                                                                   |        |                                                                                     |        |                                                                                     |        |
|                                           | 45                                                                                | 42     | 40                                                                                | 48     | 45                                                                                | 51     | 40                                                                                  | 63     | 39                                                                                  | 49     |
| Ping (milliseconds)                       |                                                                                   |        |                                                                                   |        |                                                                                   |        |                                                                                     |        |                                                                                     |        |
|                                           | 505                                                                               | 1      | 95                                                                                | 4      | 50                                                                                | 27     | 8                                                                                   | 566    | 7                                                                                   | 0      |
| File Tranfer Read and write time combined |                                                                                   |        |                                                                                   |        |                                                                                   |        |                                                                                     |        |                                                                                     |        |
| 100MB assorted                            | 40.79                                                                             | 32.49  | 30.88                                                                             | 61.81  | 28.24                                                                             | 34.63  | 24.71                                                                               | 161.01 | 42.31                                                                               | 38.63  |
| 100MB sequential                          | 46.94                                                                             | 43.2   | 30.39                                                                             | 39.68  | 25.57                                                                             | 35.6   | 21.63                                                                               | 343.42 | 36.61                                                                               | 37.89  |

was with the ASUS RT-N12HP which seemed to be lacking a bit of thought in the design. That router featured ridiculously long and heavy antennae (9dBi), but the router wasn't weighted to compensate for the high centre of gravity and we ended up tipping the router over every time we wanted to adjust the antennae.

Having external antennae is a major factor when deciding on a router as you get the freedom to orient them in the direction you prefer. You can also extend the antennae using SMA connectors and even hide your routers somewhere with antennae poking out at just the right spots. Routers with internal antennae are good if

for some reason, you're considering their aesthetical value.

## Performance

An improvement in throughput was expected due to the ac specifications and most routers didn't disappoint though there were a few that didn't perform as expected in Zone 1 as well as Zone 2. There were some basic

authentication issues when the routers were set to open authentication which is quite the opposite of what should have been. Regardless of authentication, most routers we tested didn't have a steady rate of data streaming. The graph seemed to be on a roller coaster ride with frequent spikes every now and then. There were two